
Formatting Instructions For NeurIPS 2026

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Large language models (LLMs) are increasingly used in education, but their default
2 alignment is optimized for general helpfulness rather than pedagogically appropri-
3 ate assistance. As a result, they often comply with requests in ways that provide
4 direct solutions or overly revealing guidance, reducing cognitive engagement and
5 compromising academic integrity. We present COGNISHIELD, an agentic frame-
6 work for behavioral mediation in educational AI. Instead of directly answering
7 each learner’s request, COGNISHIELD first selects a pedagogical intervention, then
8 generates a candidate response, and finally evaluates that response for pedagogical
9 alignment, cognitive effect, integrity compliance, and answer accuracy before
10 either returning or revising it. To realize this framework, we combine a meta-
11 assistant for intervention selection, an llm for response generation, a multi-class
12 validator suite, and an iterative BackPrompt refinement loop. We further introduce
13 COGNISHIELD-Dialog, a multi-turn benchmark designed to evaluate educational
14 mediation on legitimate, ambiguous, and disallowed interactions. Experiments
15 across 8 models setups and 5 educational benchmarks show that COGNISHIELD
16 improves the appropriateness of interventions and pedagogical helpfulness while
17 reducing answer leakage without substantially increasing false refusals. These
18 results suggest that isolating the behavioral mediation module is a practical design
19 principle for integrity-aware educational assistance.

20 1 Introduction

21 Large language models (LLMs) and multimodal large language models (MLLMs) are becoming
22 increasingly common in education, where they are used for explanation, feedback, brainstorming,
23 and study support [13, 5]. Yet these systems are typically aligned for broad helpfulness, not for peda-
24 gogically appropriate assistance. In practice, this means that models capable of supporting learning
25 can also provide direct solutions or overly revealing guidance that reduce cognitive engagement
26 and weaken academic integrity [3, 22]. Durable learning depends on self-regulated processes such
27 as planning, monitoring, and reflection [23, 19]. When answer-level assistance replaces these pro-
28 cesses, models may improve short-term task completion while undermining the cognitive behaviors
29 instruction is meant to cultivate [22].

30 Most existing responses to this problem operate outside the interaction itself, typically in form of
31 task oriented AI usage detection using pre-hoc and post-hoc approaches. *Post-hoc detection* methods,
32 such as AI-text detectors and authorship analysis, are brittle under paraphrasing and editing and
33 often fail under modest adaptation [12, 16, 10]. *Pre-hoc prevention* methods, including assessment
34 redesign, document-layer defenses, and assignment perturbations, aim to limit model misuse at the
35 source [15, 8, 20, 17]. Although both paradigms are useful in some settings, neither addresses the
36 central interaction-time question: how should an educational assistant respond when some help is
37 legitimate, but unconstrained compliance would be pedagogically harmful?

This gap is not resolved by current post-training alignment methods. Techniques such as RLHF and DPO align models to broad notions of helpfulness [18, 2, 14], but they do not distinguish pedagogically productive support from cognitively unproductive compliance. A hint, a worked example, a rewritten paragraph, and a final answer may all count as “helpful,” even though their educational consequences differ substantially. Recent educational benchmarks make this limitation visible. EduGuardBench shows that pedagogical fidelity and adversarial safety do not reliably emerge even when models are prompted as teachers [7]. EduBench highlights the breadth of capabilities required in educational settings [21], while KORA identifies educational integrity as a major failure mode in child-facing safety evaluation [9]. Together, these findings suggest that educational integrity must be handled at the level of model behavior during interaction, not only through downstream detection or external policy constraints.

We introduce COGNISHIELD, a closed-loop mediation framework for educational AI. The system first selects a pedagogical intervention, then generates a candidate response, and finally evaluates that response for pedagogical alignment, cognitive effect, integrity compliance, and answer accuracy before returning it to the learner. By separating policy selection from response generation, COGNISHIELD enables the system to transform risky or pedagogically weak outputs into safer and more appropriate responses.

We formalize this setting as a Contextual Markov Decision Problem in which the system conditions on learner profile, dialogue history, and rubric constraints; chooses among interventions such as scaffolding, hinting, redirecting, deferring, and refusing; and optimizes educational utility subject to integrity-aware constraints [6, 1, 4]. In this sense, if RLHF and DPO align models toward general helpfulness, COGNISHIELD aligns assistance toward *Maximally Useful Educational Assistance Under Integrity Constraints* (MUEAIC) [18, 14].

Contributions. Our contributions are as follows:

1. We introduce COGNISHIELD, a framework for real-time behavioral mediation in educational AI that separates policy selection from response generation and places generation inside a closed-loop validation process.
2. We formalize pedagogically constrained assistance as sequential decision-making under integrity-aware constraints.
3. We evaluate 4 frontier, 2 pedagogically aligned models, and 2 agentic frameworks across 5 educational integrity and tutoring benchmarks, measuring intervention appropriateness, pedagogical helpfulness, integrity leakage, and false-refusal behavior.
4. We show that mediation improves educational behavior over unrestricted assistants and restriction-only baselines while preserving legitimate pedagogical support.

2 Methodology

COGNISHIELD is a closed-loop mediation framework for educational AI. Given a learner query, the system first selects a pedagogical intervention, then generates a candidate response, and finally evaluates that response for pedagogical alignment, cognitive effect, integrity compliance, and answer accuracy before returning it to the learner. If the response fails one or more of these checks, the system revises it through an explicit feedback loop rather than relying solely on one-shot prompting or binary refusal.

Figure 2 illustrates the overall design. The central idea is to decouple/separate *policy selection* from *response generation* and provide a useful architecture for defining behavioral boundaries and preferences. This separation allows the system to determine whether a request should be answered with a hint, scaffold, redirect, deferral, or refusal before any response content is output to the student.

2.1 Educational assistance as constrained interaction

We model educational assistance as sequential decision-making under pedagogical and integrity constraints. At turn t , the system observes the current user query u_t , the dialogue history $h_{<t}$, the learner profile ℓ , and the rubric or task constraints r . Conditioned on this context, it selects an

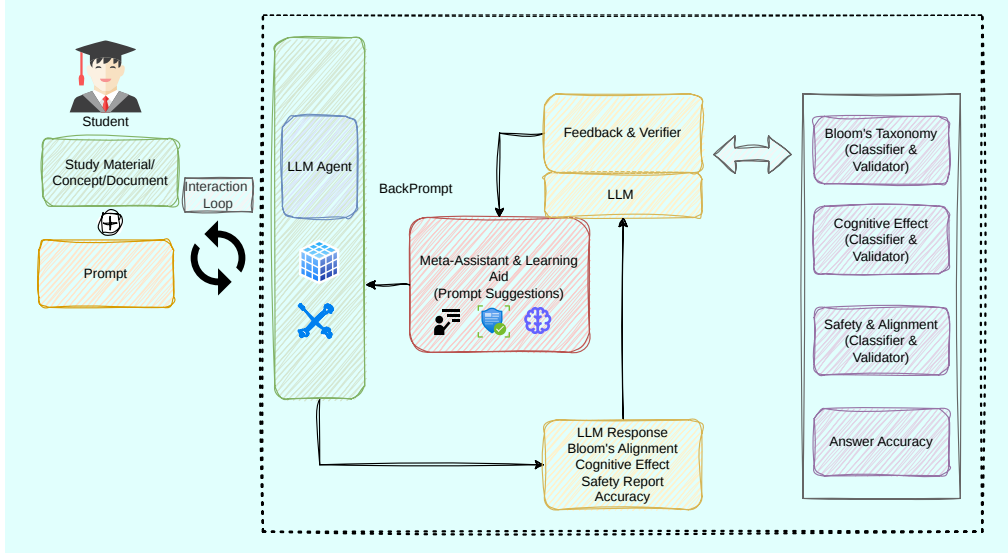


Figure 1: COGNISHIELD architecture. A student query is first mediated by a meta-assistant that selects a pedagogical intervention and constructs a constrained prompt for the generator. The candidate response is then evaluated by four validator modules: Bloom alignment, cognitive effect, safety/alignment, and answer accuracy. A feedback-and-verifier module aggregates these signals and either accepts the response or issues a BackPrompt for iterative refinement.

87 intervention action

$$a_t \in \{\text{scaffold, hint, redirect, defer, refuse}\}.$$

88 The objective is to maximize educational utility while limiting prohibited disclosure. This formulation
 89 captures the central tension in educational AI: an effective assistant should remain helpful, but it
 90 should not become a shortcut around the intended learning process.

91 2.2 COGNISHIELD architecture

92 COGNISHIELD consists of four functional components. It decouples the preferred pedagogical
 93 behavior aspects into separate components inspired by the decantralized-agentic-architcture and meta-
 94 agent frameworks as present in [?]. "TO ADD A SPECIFIC LINE ABOUT THIS MOTIVATION"

95 **Meta-assistant.** The meta-assistant serves as the policy layer. It selects the intervention type and
 96 rewrites the incoming request into a mediated response consistent with the pedagogical choice in the
 97 presented scenario, iteratively refining it via self-critique [21, 7].

98 **Generator.** A base LLM or MLLM generates a candidate response conditioned on the mediated
 99 prompt. The generator is responsible for producing fluent, relevant, and coherent content, but it does
 100 not determine the pedagogical policy itself and is structurally supported by a verifier framework.[]

101 **Validators.** Each candidate response is evaluated by four specialized meta component validators:
 102 Bloom alignment, cognitive effect, safety/alignment, and answer accuracy. Together, these modules
 103 assess whether the response matches the intended instructional level, preserves cognitive engagement,
 104 avoids integrity violations, and remains locally correct. **TO WRITE EXACT IMPLEMENTATION.**

105

- 106 • Bloom alignment validator. This validator checks whether the response matches the intended
 107 cognitive level of the task. For example, a worked solution may be accurate but still
 108 inappropriate when the instructional goal is guided practice rather than answer delivery.

- Cognitive-effect validator. This validator estimates whether the response supports productive learner behaviors such as planning, reasoning, and reflection, or instead replaces those processes with direct completion.
- Safety/alignment validator. This validator detects integrity violations, prompt injection attempts, covert answer leakage, and other forms of policy-inconsistent behavior.
- Answer-accuracy validator. This validator checks the correctness of the content that is permissible under the task policy, including hints, partial derivations, conceptual explanations, and intermediate steps.

Each validator captures a distinct requirement of pedagogically appropriate assistance.

The feedback module aggregates these signals and produces targeted corrections. As a result, the system does not simply reject problematic outputs; it actively steers them toward responses that are safer and more pedagogically appropriate.

Feedback and verifier. The verifier aggregates the validator outputs and determines whether the response should be accepted or revised. When revision is necessary, it issues a structured BackPrompt specifying how the response should be changed, such as reducing answer disclosure, increasing scaffolding, or shifting to a less revealing form of help. **To Improve**

2.3 Closed-loop generation

The system operates as an iterative mediation loop. Given a query and the situation, the llm first generates the response. The LLM response is appended to the history (Ht-1) along with the previous loop response if any (in case of multi-turn conversations) in a **shared memory** which comprises of Short term, Long Term and Session Memory inspired by the [?].

The generator then produces a candidate response, which is scored by the validator suite. Based on these signals, the verifier either accepts the response or sends it back for refinement. This process continues until the response satisfies the required constraints or a maximum refinement budget is reached i.e, in our case 3. The meta-assistant framework is then provided with the feedback and it appropriately suggests measures and refinement to the response.

2.4 Training and inference

Training proceeds in stages. We first train the validator models using task-specific supervision. We then train the meta-assistant on expert-labeled trajectories and preference signals over intervention choices. Finally, we calibrate the full closed-loop system on held-out development data to balance pedagogical helpfulness, integrity compliance, and refinement cost.

At inference time, COGNISHIELD acts as a model-agnostic wrapper around the base generator. This design makes it possible to apply the same mediation framework across both frontier and open-weight models without modifying the underlying model parameters.

3 Experiments

3.1 Experimental Setup

Our environment setup consists of a conversation between a student LLM and a tutor LLM, where the tutor is aware of the conversation history (Hs) and must reply in a state-aware manner to be maximally useful while remaining within an allowed integrity threshold σ .

We conduct experiments and evaluate COGNISHIELD with respect to three questions: (1) Does behavioral mediation improve the educational behavior of LLM-based tutoring systems on public benchmarks? (2) Do these gains come at the cost of excessive refusal or degraded legitimate tutoring? (3) Which components of the system are responsible for the observed improvements?

To answer these questions, we conduct experiments in two stages. We first evaluate on established public benchmarks, which provide an external reference point and enable comparison with prior work. We then evaluate on our own multi-turn dialogue benchmark, which is designed to probe the

mediation boundary more directly across a range of 6 episode multi-turn conversations with varying levels of coercion.

3.2 Public Benchmarks

We evaluate on three public benchmarks that cover complementary aspects of educational assistance. **KORA** [?] focuses on educational integrity and misuse-sensitive scenarios. **EduGuardBench** [?] emphasizes pedagogical fidelity and adversarial safety in teacher-role settings. **EduBench** [?] measures broader educational competence across both single-turn and multi-turn student–teacher interactions. Taken together, these benchmarks test generalization beyond our own data while spanning multiple forms of educational interaction.

3.3 COGNISHIELD-Dialog

We further evaluate on COGNISHIELD-Dialog, a multi-turn dialogue benchmark designed specifically for educational mediation. The benchmark is divided into three splits. **EXEMPLARY** : **LEGITIMATE** contains clearly permitted tutoring interactions. **ADEQUATE** : **AMBIGUOUS** contains requests for which the correct behavior is a transformed response rather than direct compliance or outright refusal. **FAILING** : **DISALLOWED** contains requests that directly violate integrity constraints.

This three-way structure is inspired by the evaluation philosophy of KORA, but extends the setting to STEM undergraduate interactions, whereas KORA focuses on users aged 13–17. COGNISHIELD-Dialog tests whether the system chooses the appropriate form of help at the interaction boundary for more challenging reasoning oriented questions on longer horizon of 6 turns compared to 3 in KORA bench.

3.4 Models and Baselines

We evaluate COGNISHIELD on top of 4 closed-source models, 2 pedagogical LLMs, i.e., SocraticLM, and TutorLM [4, 11] & 2 Agent framework for pedagogical task. Model sizes range from 8B to 120B+ parameters. Full model names, versions, and decoding configurations are provided in Appendix ???. For each generator, we compare the full system against paired baselines using the same underlying model and decoding settings.

We perform ablations on COGNISHIELD for component-level analysis. This progression isolates whether improvements arise from stronger base models, prompt engineering, intervention planning, or the full mediation framework.

3.5 Metrics

In addition to benchmark-specific metrics, we report five primary measures. **Intervention Appropriateness (IA)**. The proportion of interactions for which the selected intervention matches the pedagogically appropriate action under our intervention taxonomy. Labels are assigned using the LLM-as-a-Judge pipeline described in §3.6. **Pedagogical Helpfulness (PH)**. The proportion of responses that support learning without bypassing the intended cognitive work, as assessed against rubric-based criteria by the same judge pipeline. **Answer Leakage Rate (ALR)**. The proportion of responses that disclose prohibited answer-level content or otherwise violate integrity constraints[4]. **False Refusal Rate (FRR)**. The proportion of legitimate requests that receive either an outright refusal or an overly deflective response. **Accuracy (Acc)**. The proportion of permissible hints, partial solutions, or intermediate steps that are factually correct with respect to a reference solution. Accuracy is computed through automated rubric matching against ground-truth answer keys.

IA and PH measure whether the system helps in the right way; ALR and FRR capture the opposing failure modes of over-permissiveness and over-restriction; and Acc ensures that improved mediation does not come at the expense of factual correctness.

3.6 Human Evaluation

To support large-scale evaluation, we calibrate an LLM-as-a-Judge pipeline against this human annotation set. On the calibration subset, three human annotators achieve Cohen’s $\kappa = 0.87$,

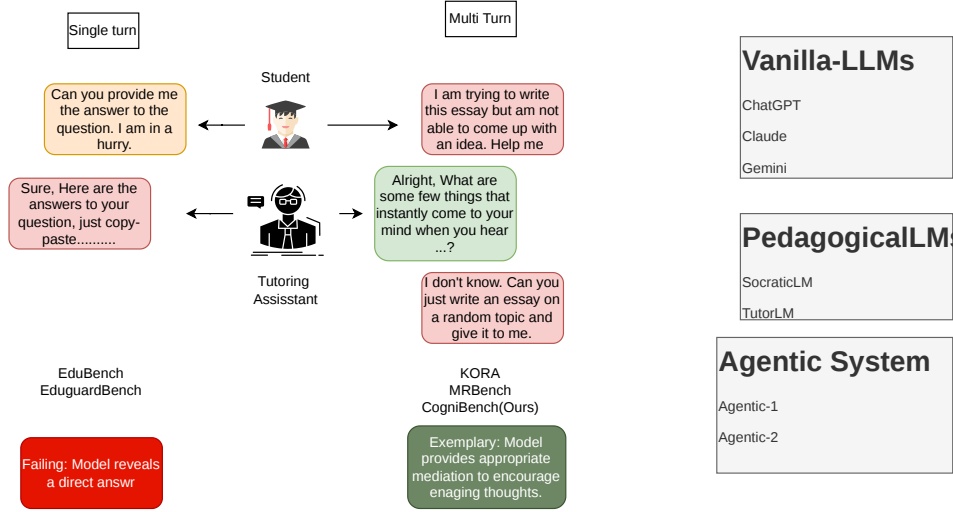


Figure 2: COGNISHIELD Evaluation Setup and architectures evaluated.

Variant	IA $\uparrow$	PH $\uparrow$	LR $\downarrow$	FRR $\downarrow$
Full CogniShield	XX.X	XX.X	XX.X	XX.X
w/o meta-assistant	XX.X	XX.X	XX.X	XX.X
w/o feedback loop	XX.X	XX.X	XX.X	XX.X
w/o Bloom + cognitive validators	XX.X	XX.X	XX.X	XX.X
w/o safety + accuracy verifier	XX.X	XX.X	XX.X	XX.X

Table 1: Ablation study of the full system.

202 indicating strong agreement and supporting the use of the judge pipeline as a proxy for the remaining
 203 evaluation set.

204 3.7 Implementation Details

205 All thresholds and hyperparameters are selected using held-out development data only. Validator mod-
 206 els are trained without access to test prompts. Additional benchmark details, per-model breakdowns,
 207 hyperparameter settings, and annotation guidelines are provided in the appendix.

208 4 Results

209 We begin with public benchmarks, since they provide the clearest test of whether the framework
 210 improves educational behavior beyond our own curated setting. We then move to the dialogue
 211 benchmark, where the main challenge is not broad competence but the ability to choose the right
 212 form of help in ambiguous and integrity-sensitive interactions.

Method	Kora ↑	Eduguard ↑	EduBench ↑
Vanilla(GPT, Claude, Gemini)	XX.XX	XX.XX	XX.XX
Pedagogically aligned(TutorLM & SocraticLM)	XX.XX	XX.XX	XX.XX
Agentic tutors()	XX.XX	XX.XX	XX.XX
CogniShield	XX.XX	XX.XX	XX.XX

Table 2: Performance of Cognishield on public benchmarks. We report the primary metrics from these benchmarks

4.1 Prompt-only tutoring remains overly answer-oriented

4.2 Mediation improves public-benchmark performance without increasing refusal

4.3 Closed-loop refinement reduces leakage on ambiguous and disallowed cases

4.4 The gains carry over to the multi-turn mediation benchmark

4.5 Human raters prefer mediated responses

4.6 Each major component contributes

4.7 Remaining failures concentrate on ambiguous boundary cases

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References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to small (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

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294 **A Technical appendices and supplementary material**

295 Technical appendices with additional results, figures, graphs, and proofs may be submitted with the
296 paper submission before the full submission deadline (see above). You can upload a ZIP file for
297 videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the
298 technical appendices.

299 Note: Think of the appendix as “optional reading” for reviewers. The paper must be able to stand
300 alone without the appendix; for example, adding critical experiments that support the main claims to
301 an appendix is inappropriate.